SENTINEL AI · BFSI CONTROL TOWER

End-to-End Readiness Report

Current implementation status, automated verification, and a bounded live-browser smoke test. Findings are distinguished from assumptions and from prior audit evidence.

Report timestamp	18 October 2026, 23:42 SGT
Repository	`Mohammed-Ilyas-AI/Sential-AI` · local working tree; no commit, push, deployment or cloud operation performed for this report
Browser target	Local Angular application at `http://localhost:4200/control-tower`, shared browser, approximately 927 × 420 viewport
Automated verification	`npm run verify` completed successfully on 18 October 2026, approximately 23:38 SGT
Scope	Eight MVP modules, core control path, synthetic scenario, dashboard browser smoke check, and known launch/tenant/performance limitations

All institutions and data are synthetic. The bank lens is a presentation/projection mechanism, not authentication or production tenant isolation. No real bank integration or COSMIC integration is claimed.

1. Executive result

Automated repository verification passes. The current source tree typechecks, passes ESLint, passes the workspace test suites and builds all workspaces. The deterministic policy path and the synthetic takeover scenario are covered by tests. The Case Reports, graph, account controls, audit, performance and seeded-noise workstreams are present in the local implementation.

This is not a production deployment certification. This run did not call deployed services, operate cloud resources, invoke Gemini, or exercise every UI route in a browser. The live browser check below is limited to the local Control Tower's bank focus/reset behavior. Stream tenant filtering and several performance dimensions remain unimplemented.

387
automated tests passed

21
control-path scenario tests

8 / 8
MVP modules implemented; scenario launcher capability gated

1
browser interaction flow exercised here

2. Verification method and limits

METHOD	RESULT / SCOPE	BOUNDARY
`npm run verify`	**PASS** Typecheck, lint, all workspace tests and builds; detailed counts in section 8.	Local repository state only.
`npm run test:scenario`	**PASS** Three scenario test files, 21 tests including canonical repeatability and isolated synthetic noise.	Harness-based control path, not a deployed bank integration.
Shared browser: Control Tower	**SMOKE PASS** Page rendered. Bank B marker selection showed the focused state; Reset global view restored Singapore / All Banks and cleared the focus.	No full cross-page or multi-viewport browser regression was run.
Production, cloud and model checks	**NOT RUN**	No deployed endpoint call, Cloud Run operation, or Gemini invocation was made in this update.

3. MVP implementation status

MODULE	STATUS	CURRENT IMPLEMENTATION AND VERIFICATION EVIDENCE
1 · Transaction simulator	**Implemented**	Deterministic `DEMO_TAKEOVER_01`, run-scoped IDs, synthetic noise generator disabled by default, and bounded rate/seed configuration. Scenario suite confirms noise is isolated and does not HOLD.
2 · Real-time risk engine	**Implemented**	Seven pure rules and the decision gate remain deterministic. Rule tests, control-path API tests and the canonical scenario exercise ALLOW/HOLD behavior without any model decision.
3 · Rapid-drain detection	**Implemented**	Thresholds remain domain configuration; score contributions are distinct from the policy gate. Canonical scenario regression preserves the documented score ladder, threshold crossing and hold behavior.
4 · Intervention workflow	**Implemented**	Append-only customer contact records, account kill-switch history and customer-verified release path are wired with audit records. Control-path enforcement keeps the policy decision separate from score changes.
5 · Investigation workspace	**Implemented**	Evidence bundle and advisory endpoints use recorded evidence. Cross-lens investigation reads are denied before bundle compilation or model invocation; covered by API tests.
6 · AI explanation	**Implemented in code**	Grounded explanation and summary orchestration remain in the core AI infrastructure boundary. This report run did not invoke Gemini or remeasure live latency; do not treat past live calls as reverified here.
7 · Entity graph	**Implemented**	Bounded two-hop traversal, graph projection, fund trace, evidence provenance and tests. The implementation reuses indexed BFS rather than a single `$graphLookup`; this deviation is documented.
8 · Case report	**Implemented**	Case lifecycle, evidence-backed report generation, draft edits with revision checks, sign-off, revisions, text export and audit persistence. Only previously archived validated summaries are included; report generation does not issue a model call.

4. Control-path and scenario results

The core contract remains: **rules decide, AI explains, humans authorize.** The decision path does not call a language model; score changes do not authorize a transaction. Contributions are recorded as the score source, interventions are append-only, and lifecycle events are tested for idempotent handling.

SCENARIO CHECK	OBSERVED TEST OUTCOME
Canonical scenario score ladder	Scenario regression asserts the documented sequence and final hold score against `DEMO_TAKEOVER_01`.
Noise traffic isolation	Seeded, run-scoped events avoid the canonical victim; sampled noise transfers land in the ELEVATED band and do not HOLD. Canonical replay remains unchanged after noise playback.
Policy and intervention paths	Core API tests cover account controls, customer contact, hold cascade, graph, case reports and investigation authorization; see the full suite counts below.
Cloud execution	Not started from the browser. No unauthenticated Cloud Run Job trigger is exposed.

5. Live browser smoke check

Page: `http://localhost:4200/control-tower`. The shared browser showed the Control Tower, headline metrics and the Live Singapore Network panel with the bank markers, map controls, route legend, transaction table and system status.

ACTION	OBSERVED RESULT	STATUS
Load Control Tower	Rendered without a browser error in the shared local tab. At capture, displayed 200 live transactions, 7 high-risk alerts, 1 active hold, 3 monitored banks; network figures showed 3 banks, 12 active routes and 76 monitored accounts. These are time-specific UI readings, not invented fixtures or guaranteed current counts.	**Verified**
Select the Bank B map marker	Bank B marker became active, the bank focus selector selected Bank B, and the view indicated it was focused.	**Verified**
Click Reset global view	Selector returned to Singapore (All Banks); Show all banks returned to its pressed/global state; focused state cleared. The map returned to the all-bank view. No theme control was changed during this interaction.	**Verified**
Select Bank A, select a transaction, then reset	Not performed in this browser run.	**Not verified**
Zoom/rotation, layers, reduced motion, dark mode, narrow/large desktop	Not manually exercised in this browser run. Automated UI suite passed; that is not a substitute for visual regression at these conditions.	**Not verified**

The browser viewport was approximately 927 × 420 pixels. This check verifies the accessible bank-focus and reset controls, not pixel-perfect appearance against a separate design reference.

6. Supporting capabilities and boundaries

CAPABILITY	CURRENT STATE	IMPORTANT LIMIT
Case/report API and UI	Persisted case/report lifecycle with audited operations; report export is saved UTF-8 text.	No PDF report renderer is claimed; UI workflow was not manually walked in the browser during this verification.
Entity graph	Bounded graph API with two-hop neighbourhood and explicit fund trace behavior.	Uses indexed BFS to preserve directed provenance, not the originally proposed single MongoDB graphLookup.
Tenant lens	Projection/authorization checks cover supported core resources, including graph, cases, account controls, interventions and investigations; cross-lens investigation denial is tested and audited.	**Not authentication.** The WebSocket stream still broadcasts globally and does not filter messages per tenant connection; production tenant isolation is not established.
Performance panel	Core records fixed-label request timings in a bounded five-minute window; dashboard exposes counts, failures and latency percentiles.	WebSocket delivery, worker batches and browser rendering are not measured. Empty percentiles are null, not fabricated values.
Scenario launcher	UI now states the launch capability is unavailable.	No authenticated operator principal or configured server-side Cloud Run Job invoker exists. Simulator can run through its configured local/job entrypoint; do not add a browser-triggered unauthenticated job call.
AI summary in reports	Report may include a previously archived, validated grounded summary.	Report generation itself does not call Gemini. AI never returns a policy decision.

7. Readiness actions

1. **Keep the scenario-launch blocker explicit** until the platform provides authenticated operator identity, server-side job configuration/credentials, idempotent run persistence and truthful status polling. See `docs/decisions/ADR-010-scenario-launcher-auth-boundary.md`.
2. **Complete per-connection tenant filtering for the stream** before presenting the tenant lens as an isolation boundary. Continue labelling the current lens as presentation/projection only.
3. **Add browser E2E coverage** for bank A → bank B, transaction selection → global reset, case report workflow, layers/zoom, reduced motion, light/dark themes and responsive desktop dimensions.
4. **Add telemetry only for measured paths**; separately instrument stream delivery, worker processing and browser rendering before claiming those targets.
5. **Run deployment and Gemini verification separately** in an explicitly authorized environment. This local report does not assert cloud readiness, live model availability, or deployment status.

8. Automated test evidence

WORKSPACE / COMMAND	RESULT
`npm run typecheck` (all workspaces)	**PASS** Domain, rules, scenarios, core, stream, simulator and UI.
`npm run lint`	**PASS** ESLint including the repository import-boundary rules.
`@sentinel/domain`	10 tests passed.
`@sentinel/rules`	62 tests passed.
`@sentinel/scenarios`	21 tests passed.
`@sentinel/core`	166 tests passed, including graph/case/performance/account-controls APIs and cross-lens investigation denial.
`@sentinel/stream`	8 tests passed. These do not validate tenant-specific filtering, which is not implemented.
`@sentinel/simulator`	8 tests passed.
`@sentinel/ui`	112 tests passed.
`npm run build` (all workspaces)	**PASS** TypeScript applications and Angular production bundle built successfully.
`npm run test:scenario`	**PASS** 21 tests across 3 files.
`git diff --check`	**PASS** No whitespace errors.

Total: 387 tests passed across the full workspace run (10 + 62 + 21 + 166 + 8 + 8 + 112). Test counts are from the successful run at approximately 23:38 SGT on 18 October 2026.

9. Scope and data handling

- All scenario records and participating institutions are synthetic.
- No backend API schema, policy threshold, score curve or cloud resource was changed for this report update.
- No credentials, customer data, secrets or model identifiers were copied into this report.
- The report reflects the local source tree and the explicitly described local browser check. It does not represent a deployment approval or security certification.

SENTINEL AI · Internal E2E readiness evidence · Updated 18 October 2026, 23:42 SGT. Earlier readiness reports and other files in `docs/reports` were not used as current verification evidence.